Car Brand Recognition

Naming the make of a car from a single photo with deep learning

Deep Learning semester project

The problem: which brand is this car?

- Input: one photo of a car, any angle
- Output: the brand, from 49 makes, with a confidence score
- Hard because a brand spans many shapes, colours and years, while different brands can look alike
- Use cases: parking and tolls, listings, insurance, fleet analytics

49

brands to tell apart

16,185

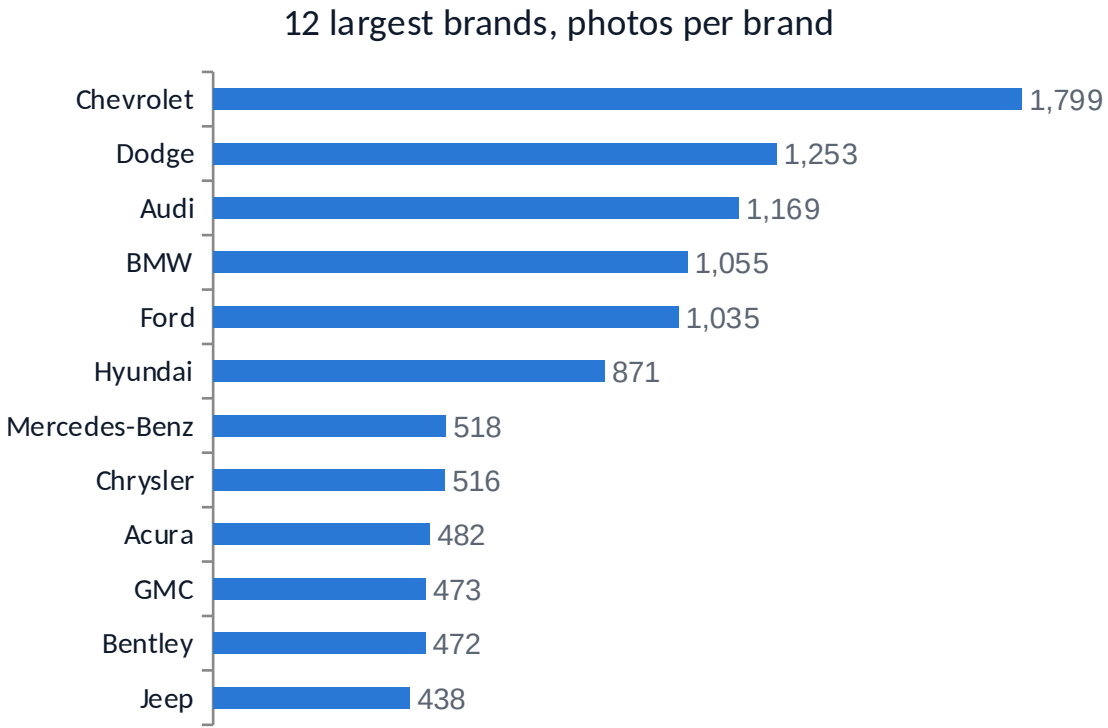
labelled photos in the dataset

Dataset: Stanford Cars

7,331
train photos

813
validation photos

8,041
test photos (official split)



The pipeline, end to end

1

Data

Stanford Cars, make extracted as label

2

Preprocess

224 x 224 RGB, train / val / test

3

Train

transfer learning + fine-tuning

4

Evaluate

accuracy, F1, confusion matrix

5

Export

Keras to ONNX, 24 MB

6

Deploy

Vercel site, runs in the browser

Model: transfer learning with EfficientNetV2-B0

Photo

224 x 224 x 3

EfficientNetV2-B0

ImageNet weights, 5.9M params

Average pool

1280 features

Dropout + Dense

softmax, 49 brands

Stage 1: linear probe

Backbone frozen, only the final layer is trained. Test accuracy 60.4%.

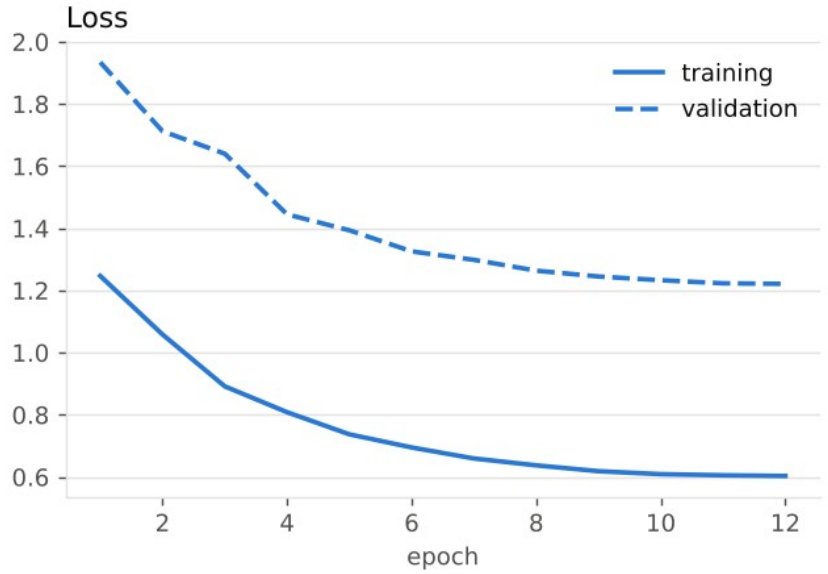
Stage 2: fine-tuning

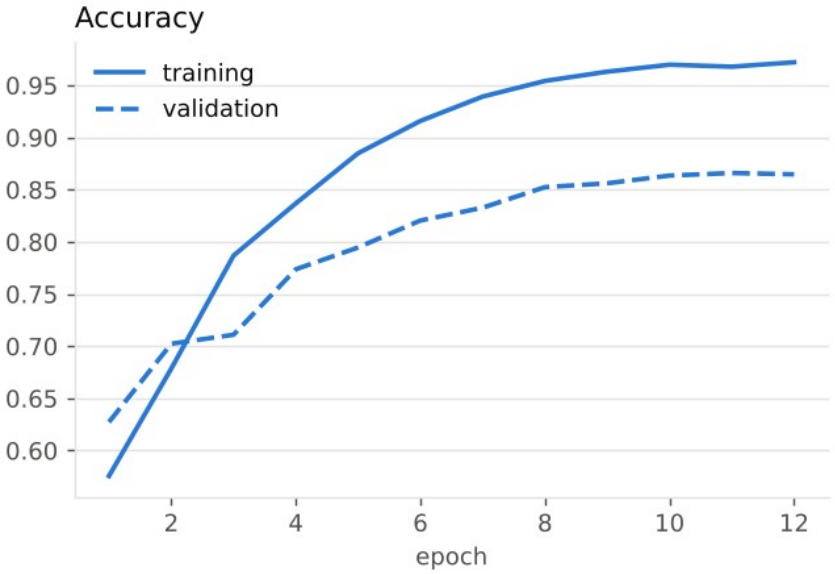
Upper blocks unfrozen and trained for 12 epochs with augmentation. Test accuracy 86.8%.

Training setup

Setting	Value
Backbone	EfficientNetV2-B0, ImageNet weights
Optimiser	AdamW, weight decay 1e-4, peak LR 2e-4, cosine decay
Loss	Cross-entropy, label smoothing 0.1, class weights
Batch / epochs	32 / 12
Augmentation	flip, rotation, shift, zoom, brightness, contrast
Model selection	best validation accuracy
Hardware	CPU only (4 cores)

Training curves





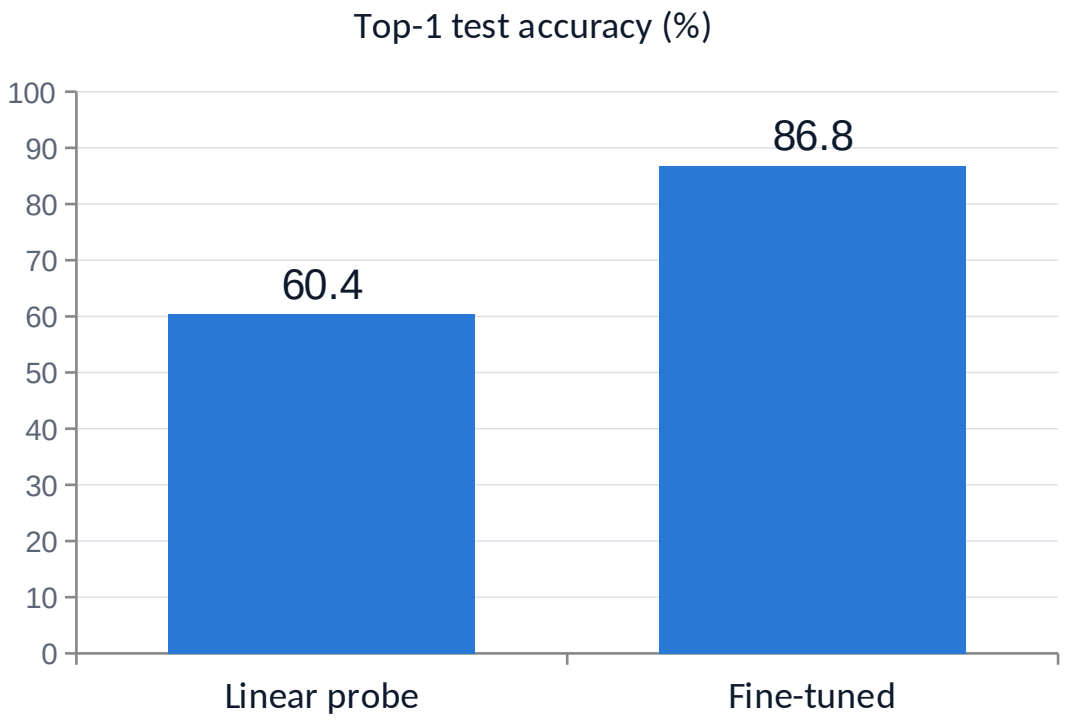
Best validation accuracy: 86.6% (813 photos, used only to choose the epoch).

Results on unseen test photos

86.8%
top-1 accuracy

95.8%
top-3 accuracy

0.87
macro F1



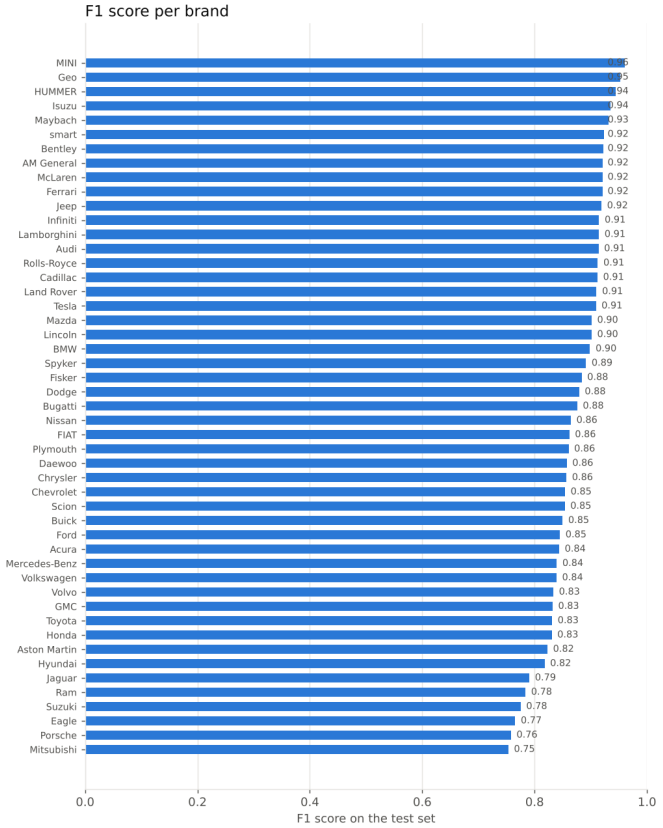
Easy and hard brands

Best

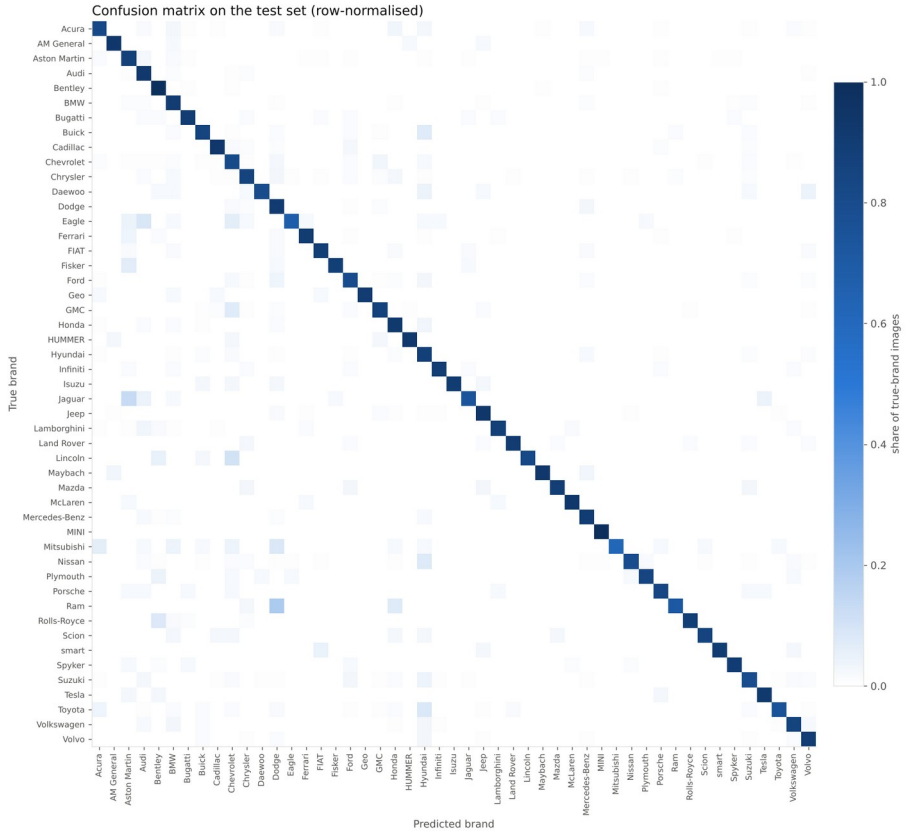
MINI 0.96
Geo 0.95
HUMMER 0.94
Isuzu 0.94
Maybach 0.93

Hardest

Mitsubishi 0.75
Porsche 0.76
Eagle 0.77
Suzuki 0.78
Ram 0.78



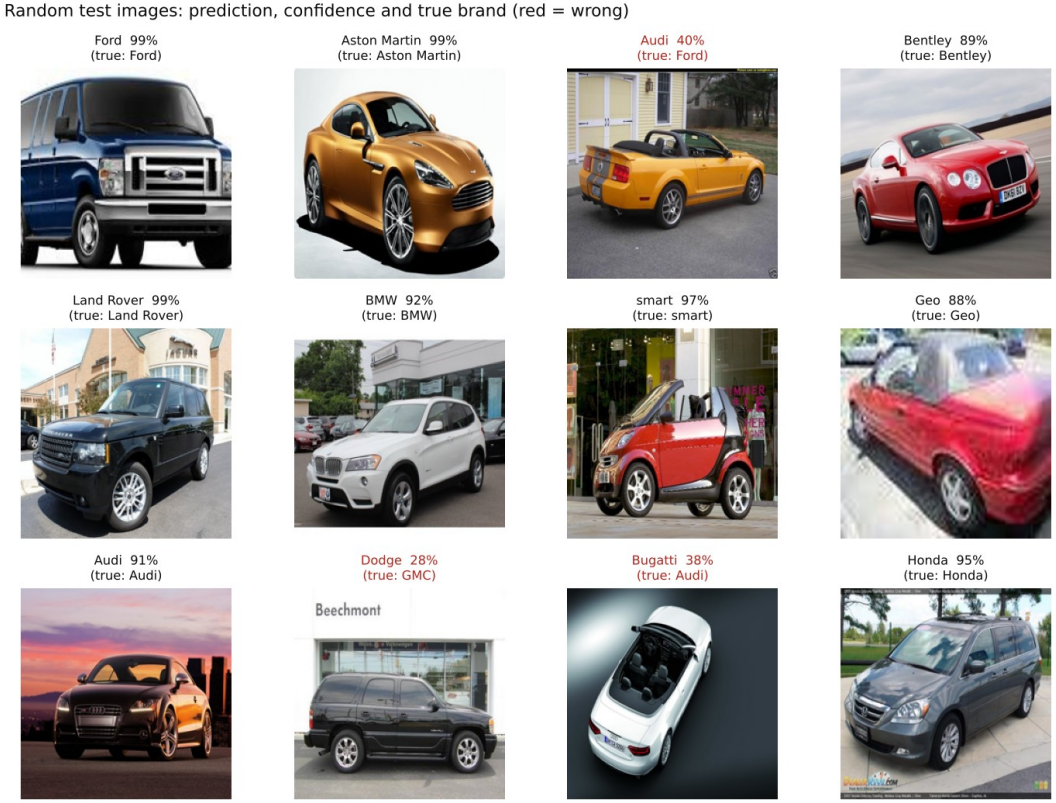
What does the model mix up?



Most frequent mix-ups

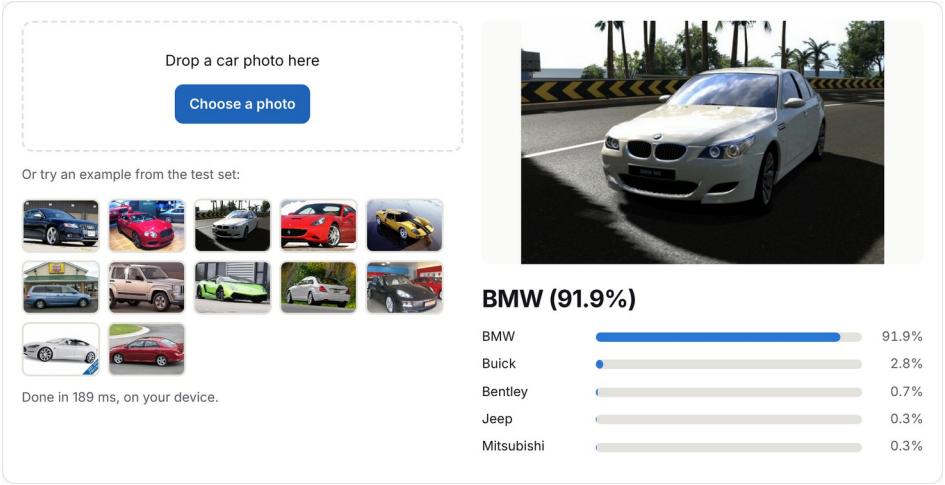
True	Predicted as	Photos
Chevrolet	GMC	31
Chevrolet	Dodge	22
Ford	Dodge	20
Chevrolet	Hyundai	18
Dodge	Mercedes-Benz	18

Sample predictions



Deployment: a website that runs the model

- Keras model exported to ONNX (24 MB), same predictions as the original
- ONNX Runtime Web runs inference in the browser, no backend
- The photo stays on the user's device
- Static site on Vercel: demo, dataset and code, results, project links



Limitations and future work

Limitations

- Some brands appear through a single model
- Mostly cars from 2000-2012
- No 'unknown' or 'not a car' class
- Small brands have few training photos

Future work

- Detect and crop the car first
- More recent, larger dataset
- Add an 'unknown' class
- Higher resolution and test-time augmentation

Conclusion

- A fine-tuned EfficientNetV2-B0 recognises 49 car brands with 86.8% top-1 and 95.8% top-3 accuracy on unseen photos.
- Fine-tuning improved the frozen-backbone baseline from 60.4% to 86.8%.
- The model runs in the browser, so the whole app is a static website.

Thank you